

MSTool-AI

Development Guidelines

Document ID
DEV-GUIDE-001

Version
1.0

Date
April 12, 2026

Classification
IEC 62304 Class C



CRITICAL — READ BEFORE CONTRIBUTING

This project is IEC 62304 Class C — the highest software safety class.
A software failure could contribute to DEATH or SERIOUS INJURY.
Every code change must follow these guidelines without exception.
Violations will result in audit findings and potential regulatory action.

1

Before Writing Any Code

1

Check the Requirement

Every change must trace to a requirement in **SRS-001** (docs/iec62304/02_Software_Requirements_Specification.md). If no requirement exists, **create one first**.

2

Check the Risk

If your change touches a Class C module (see Section 2), review **RMF-001** (docs/iec62304/03_Risk_Management_File.md) for related hazards.

3

Never Skip Code Review



Mandatory Rule

Never modify a Class C module without a formal code review — no exceptions.

2

Class C Modules

The following modules are classified as **Software Safety Class C** — the highest safety class under IEC 62304. Extra care, mandatory review, and full traceability are required for any change.

Module	Path	Function
ai_segmentation_service.py	backend/app/services/	AI-driven brain segmentation (Vertex AI proxy)
brain_volumetry_service.py	backend/app/services/	Brain structure volume computation
brain_report_service.py	backend/app/services/	Clinical report generation (Claude API)
lesion_analysis_service.py	backend/app/services/	MS lesion analysis + DIS criteria
ms_region_classifier.py	backend/app/services/	MAGNIMS region classification (PV/JC/IT/DWM)
nifti_utils.py	backend/app/utils/	NIfTI file processing + orientation
dicom_utils.py	backend/app/utils/	DICOM parsing + DICOM-SEG export
edgeAI.worker.ts	frontend/src/workers/	Browser ONNX inference (screening)

3

Pull Request Requirements

Every pull request **MUST** include:

- ☐ Description of what changed and WHY
- ☐ Reference to requirement ID (e.g., "Implements REQ-FUNC-040")
- ☐ If safety-related: reference to hazard ID (e.g., "Risk control for HAZ-002")
- ☐ All CI checks must pass before merge
- ☐ At least 1 code review approval
- ☐ For Class C changes: at least 2 code review approvals

 CI Pipeline Checks (all must be GREEN)

TypeScript compilation (npm run build)
Frontend build (npm run build)
Backend tests (python -m pytest)
SOUP vulnerability scan (npm audit + pip-audit)
Python syntax verification (ast.parse)

4 After Deploying

After every deployment, run the endpoint verification script. **All 9 checks must PASS.**

```
bash test_endpoints.sh

# Expected output:
Check 1: Health endpoint ..... PASS
Check 2: Auth login ..... PASS
Check 3: File upload ..... PASS
...
Check 9: AI segmentation ..... PASS

Result: 9/9 PASS – deployment verified
```

5 Templates & Records

Fillable PDF audit evidence forms are in docs/iec62304/templates/. Use them for every code review, test, release, and risk verification.

Template	ID	When to Use
Problem Report	TPL-01	When a bug is found
Release Checklist	TPL-02	Before every deployment
Code Review Checklist	TPL-03	For every PR review
Risk Control Verification	TPL-04	Verifying risk controls
Design Review Record	TPL-05	Architecture/design reviews
Test Execution Report	TPL-06	Testing requirements
SOUP Vulnerability Review	TPL-07	Monthly dependency check
Serious Incident Report	TPL-08	Safety incidents
Change Control Record	TPL-09	Significant code changes
Quality Gate Approval	TPL-10	Phase gate approvals
Document Approval Record	TPL-11	Formal document sign-off

6 Coding Standards

Language	Standard	Enforcement
TypeScript	ESLint rules + TypeScript strict mode	CI pipeline (npx tsc --noEmit)
Python	PEP 8 + type annotations	CI pipeline (ast.parse + syntax check)

Commit message format:

```
feat(REQ-FUNC-040): Implement volume computation

Implements voxel counting with normative percentile comparison.
Risk control RC-004 (volumetry displays percentile ranges).
Unit test: UT-VOL-001
```

7 Regulatory References

Standard	Title	Relevance
IEC 62304:2006+A1:2015	Medical device software — Software life cycle processes	Primary lifecycle standard
ISO 14971:2019	Medical devices — Application of risk management	Risk management framework
IEC 81001-5-1:2021	Health software — Security for health IT products	Cybersecurity requirements
EU MDR 2017/745	European Medical Device Regulation	Market access regulation
EU AI Act 2024/1689	Artificial Intelligence Act	AI-specific requirements